

Simple, highly reliable network communication  
system



# ISE2016D Unmanaged Industrial Ethernet Switch

· Plug-and-play

· DIN-rail

· Fanless

· IP30

# 1. Product Overview

The ISE2016D is designed for power, transportation, industrial control, and demanding industrial environments.

## Product Features:

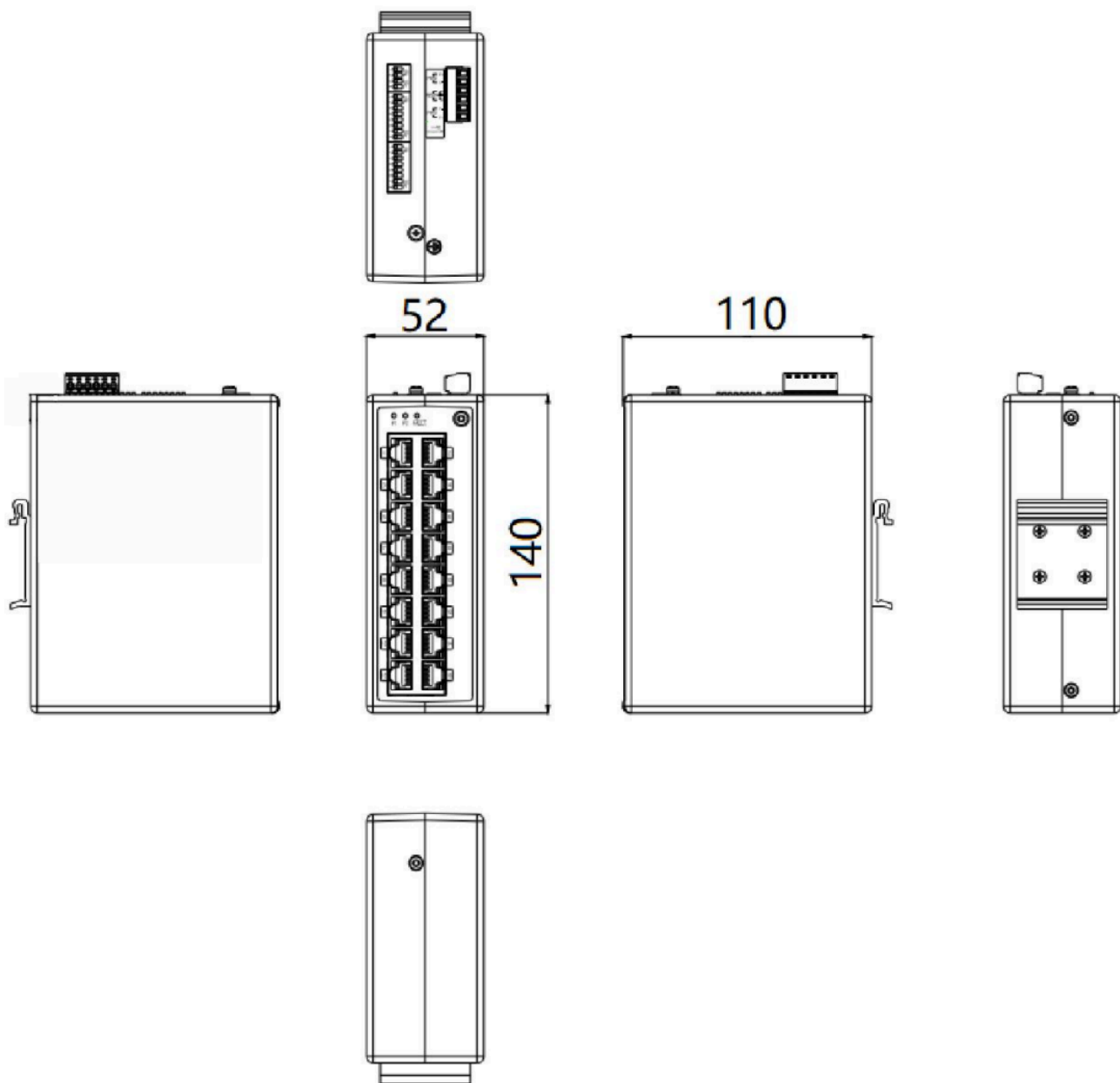
- **Robust Design:** Fully enclosed seamless metal casing, fanless design, IP30 protection rating.
- **Wide Adaptability:** Wide-temperature (-40 ~ +75°C) and wide-voltage (9.6-60VDC & 18-30VAC) design.
- **International Standard:** Complies with FCC, CE, ROHS, and UL standards with excellent EMC.
- **Smart Switching:** QoS, BSP, Power Alarm, Port Alarm and Alarm Buzzer, configurable via DIP switch.
- **Easy Deployment:** Plug-and-play, DIN rail mountable, compact size for quick deployment.

## Key Technical Specifications

| Parameter             | Specification                                                       |
|-----------------------|---------------------------------------------------------------------|
| Type                  | Unmanaged Industrial Ethernet Switch, Layer 2, Store-and-Forward    |
| Ports                 | 16 x 10/100 BaseT                                                   |
| Switching Performance | 8.8 Gbps backplane bandwidth, 8K MAC, 4 Mbit buffer, <10 us latency |
| Dimensions / Weight   | 52 x 140 x 110 mm / 1 kg                                            |
| Power                 | 9.6-60 VDC & 18-30 VAC, redundant dual inputs, 10 W                 |
| Environment           | -40 to +75 C operating, IP30, fanless                               |

| Parameter      | Specification                         |
|----------------|---------------------------------------|
| EMC            | IEC 61000-4-2/3/4/5/6/8/18, Class 3-5 |
| Certifications | CE, FCC, UL                           |

## 2. Product Dimensions



ISE2016D

Note:

1. All dimensions are in millimeters (mm).

2. All dimensions are approximate and **for reference only**.
3. The dimensions shown in the figure **shall not be used for production or processing**.
4. Dimensions must comply with part and manufacturing tolerance requirements.
5. Dimensions are subject to change without notice.

### 3. Hardware Specifications

| Category/Parameter          | Specification                           |
|-----------------------------|-----------------------------------------|
| <b>Physical Performance</b> |                                         |
| Enclosure                   | Fully enclosed seamless metal enclosure |
| Dimensions (W × D × H)      | 52 mm × 140 mm × 110 mm                 |
| Weight                      | 1 kg                                    |
| Mounting Method             | DIN-rail mounting                       |
| Cooling Method              | Fanless cooling                         |
| Ingress Protection          | IP30                                    |
| Storage Temperature         | -40 °C ~ +85 °C                         |
| Operating Temperature       | -40 °C ~ +75 °C                         |
| Humidity                    | 5 ~ 95% (non-condensing)                |
| <b>Hardware Performance</b> |                                         |
| Backplane Bandwidth         | 8.8 Gbps                                |

| Category/Parameter                     | Specification                                                                                                                                                                                      |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Processing Type                        | Store-and-Forward                                                                                                                                                                                  |
| MAC Table Size                         | 8K                                                                                                                                                                                                 |
| Packet Buffer Size                     | 4 Mbits                                                                                                                                                                                            |
| Switching Delay                        | <10 $\mu$ s                                                                                                                                                                                        |
| DIP Switch                             | Quality of Service (QoS), Broadcast Storm Protection (BSP), Power Alarm, Port Alarm, Alarm Buzzer                                                                                                  |
| <b>Power Parameters</b>                |                                                                                                                                                                                                    |
| Operating Voltage                      | 9.6–60 VDC & 18–30 VAC, Redundant dual inputs                                                                                                                                                      |
| Power Consumption                      | 10 W                                                                                                                                                                                               |
| Overcurrent Protection                 | Supported                                                                                                                                                                                          |
| Reverse Polarity Protection            | Supported                                                                                                                                                                                          |
| <b>Electromagnetic Characteristics</b> |                                                                                                                                                                                                    |
| EMS                                    | IEC(EN)61000-4-2, Class 3; IEC(EN)61000-4-3, Class 3<br>IEC(EN)61000-4-4, Class 3; IEC(EN)61000-4-5, Class 3<br>IEC(EN)61000-4-6, Class 3; IEC(EN)61000-4-8, Class 5<br>IEC(EN)61000-4-18, Class 3 |
| <b>Certifications</b>                  |                                                                                                                                                                                                    |
| Certifications                         | CE, FCC, UL                                                                                                                                                                                        |
| <b>Quality Assurance</b>               |                                                                                                                                                                                                    |
| Warranty Period                        | 5 years                                                                                                                                                                                            |
| MTBF                                   | 35 years                                                                                                                                                                                           |

## DIP Switch Settings

| Category/Parameter                     | Specification                                                                                                                                                |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Quality of Service (QoS) – On          | Enables the QoS function, processes packet priorities in 2 WRR queues. Priority 7,6,5,4 → Queue 1 (WRR Weight 16); Priority 3,2,1,0 → Queue 0 (WRR Weight 1) |
| Quality of Service (QoS) – Off         | Disables the QoS function                                                                                                                                    |
| Broadcast Storm Protection (BSP) – On  | Enables broadcast storm protection for all ports (allows max 200 broadcast packets per second)                                                               |
| Broadcast Storm Protection (BSP) – Off | Disables the broadcast storm protection function                                                                                                             |
| Power Alarm – On                       | Enables power alarm output when power supply fails                                                                                                           |
| Power Alarm – Off                      | Disables the power alarm function                                                                                                                            |
| Port Alarm – On                        | Enables port alarm output when port link fails                                                                                                               |
| Port Alarm – Off                       | Disables the port alarm function                                                                                                                             |
| Alarm Buzzer – On                      | Enables audible alarm buzzer for alarm events                                                                                                                |
| Alarm Buzzer – Off                     | Disables the alarm buzzer                                                                                                                                    |

## 4. Ordering Guide

| Model             | Description                                                                                                                                                                                                                                                                                                                    |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ISE2016D-P-16T-24 | 16-port Layer 2 unmanaged Industrial Ethernet Switch. 16 × 10/100 BaseT Ports. DIP-enabled Sound & Light Alarm Output & 1 Alarm Relay port (1A@24VDC). IP30 Protection Class, Operating Temperature from -40°C to +75°C. Dual 12/24/48 VDC (9.6~60VDC) & 24VAC (18~30 VAC) Power Input (Only connect to class 2 power supply). |

## 5. Contact Us

- Website: [InHand Networks](http://InHand Networks)
- Copyright: ©InHand Networks All rights reserved